

Hissing and Hushing in English Taboo Words

A Quantitative Analysis of Sibilant Density in English Taboo Vocabulary

1. Background

The relationship between the sound of a word and its meaning is one of the foundational puzzles of linguistics. Ferdinand de Saussure famously argued that this relationship is essentially arbitrary: there is no inherent reason that the sound sequence /dɒg/ should refer to a four-legged mammal rather than anything else. For most of the twentieth century, arbitrariness was treated as a near-universal property of human language. Over the last several decades, however, growing evidence from sound symbolism research has shown that the arbitrariness assumption is not absolute, and that certain sound–meaning correspondences recur across languages and emerge reliably in psycholinguistic experiments (Svantesson, 2017).

Taboo language offers an interesting test case for the sound symbolism hypothesis. Taboo words - profanity, slurs, and other lexical items subject to social censorship - are known to carry heightened emotional salience: they are processed faster than neutral words in lexical decision tasks, elicit measurable physiological responses, and are recalled more reliably in memory paradigms (Reilly et al., 2020). The question of whether this emotional salience is reflected in the form of taboo words, rather than only in their meanings, is a natural extension of sound symbolism research into one of the most culturally specific corners of the lexicon.

A direct empirical bridge to this question is provided by Aryani, Conrad, Schmidtke, and Jacobs (2018), who showed that words containing hissing sibilants raise the acoustic spectral center of gravity of a word and are rated as more arousing and more negatively valenced by native speakers. The authors also observed that all seven of George Carlin's famous "words you can't say on television" contain either voiceless stops or sibilants, suggesting that the phonological structure of taboo language may not be arbitrary at all. Their study, however, focused on the

acoustic and perceptual side of the question -it did not directly test whether the English taboo lexicon statistically exploits this pattern.

The present study tests this lexical-statistical prediction. Specifically, the research question is: *to what extent do sibilant phonemes (/s/, /z/, /ʃ/, /ʒ/, /tʃ/, /dʒ/) occur more frequently in English taboo words compared to a frequency- and length-matched neutral lexical baseline?* The hypothesis is that English taboo words contain a significantly higher density of sibilant phonemes than neutral vocabulary matched for word frequency and length. The null hypothesis predicts no difference between the two groups.

2. Method

2.1 Taboo word stimuli

A curated list of 29 single-word English taboo items was assembled, spanning three broad categories: sexual and anatomical terms, general profanity and identity-based slurs. Item selection was informed by the inventories of established taboo word norming studies, particularly Janschewitz (2008) and Sulpizio et al. (2024), the latter of which uses a community-based nomination procedure that improves on earlier expert-curated lists. Only single-word base forms were included, excluding compounds, multi-word expressions, and morphologically inflected variants in order to avoid double-counting the same root. Emotionally heavy but non-censored words such as *death*, *suicide*, or *moron* were excluded, since they do not satisfy a working definition of taboo as language subject to social or institutional censorship.

2.2 Phonemic transcription

Each taboo word was transcribed into its phonemic representation using the Carnegie Mellon University Pronouncing Dictionary (CMUdict), accessed through Python's nltk library (Weide, 1998). CMUdict provides ARPAbet phoneme transcriptions for approximately 134,000 English words. Phonemic transcription is essential because English orthography is unreliable: the words *sure*, *rose*, and *measure* all contain sibilants that simple letter-counting would either miss or misclassify. Of the 29 taboo items, 24 (83%) were found directly in CMUdict. The remaining five — *twat*, *dickhead*, *tits*, *bollocks*, and *wanker* — were transcribed manually using standard

pronunciation references and converted to ARPAbet. All manual additions are documented in the analysis script.

2.3 Sibilant density measure

Sibilants were defined as the six-member ARPAbet set {S, Z, SH, ZH, CH, JH}, corresponding to the IPA phonemes /s/, /z/, /ʃ/, /ʒ/, /tʃ/, and /dʒ/. For each word, the number of sibilant phonemes was counted and divided by the total phoneme count to compute sibilant density, a normalized score between 0 and 1. This normalization controls for word length, allowing direct comparison between short and long words. Vowel stress markers (the digits in ARPAbet symbols such as AH1, EH0) were stripped before counting, since these are positional annotations rather than phoneme distinctions.

2.4 Neutral baseline construction

A pool of affectively neutral candidate words was constructed using two peer-reviewed psycholinguistic databases. Warriner, Kuperman, and Brysbaert (2013) provide affective ratings for 13,915 English lemmas, including mean valence (1 = very negative, 9 = very positive) and arousal (1 = calm, 9 = excited) on a 1–9 scale. Neutral was operationally defined as valence between 4.5 and 5.5 (the true middle of the scale) and arousal below 4.5. This filter excludes both positively and negatively valenced words as well as emotionally arousing items, leaving 3,522 affectively flat lemmas. SUBTLEX-US (Brysbaert & New, 2009), a frequency corpus derived from film and television subtitles for 74,286 words, provided log₁₀ word frequency (Lg10WF) values. After intersecting Warriner-neutral words with SUBTLEX and CMUdict (ensuring frequency and pronunciation data were both available), the final neutral candidate pool contained 3,375 words. Taboo items that might have slipped through the affective filter were excluded as a safety check.

2.5 Matching procedure

For each of the 29 taboo words, a single neutral word was selected from the candidate pool using a greedy nearest-neighbor matching algorithm. For each taboo target, the standardized Euclidean distance to every available neutral candidate was computed over two dimensions - phoneme length and log-frequency - and the closest candidate was selected. The distance formula weighted a difference of two phonemes as roughly equivalent to a difference of one

log-frequency unit. Once a neutral word was selected, it was removed from the pool so no neutral item served as a control for more than one taboo word. Post hoc verification confirmed the matching was successful: mean phoneme length was 3.86 in both groups (identical to two decimal places), and mean log-frequency was 2.83 (taboo) versus 2.82 (neutral). Any observed difference in sibilant density between the groups can therefore be attributed neither to length nor to frequency confounds.

2.6 Statistical analysis

Because each taboo word was paired with a specific matched neutral word, paired statistical tests were used rather than independent-samples tests, which is both methodologically appropriate and statistically more powerful. The Shapiro-Wilk test was applied first to assess normality. Given the bounded nature of sibilant density (most words contained zero sibilants), the non-parametric Wilcoxon signed-rank test was selected as the primary analysis, with the paired t-test reported for parametric comparison. Both tests were one-tailed, consistent with the directional hypothesis that taboo words contain *more* sibilants than neutrals. Cohen's *d* for paired samples was computed to report effect size independently of statistical significance. All analyses were performed in Python 3.14 using pandas, scipy, and nltk; the full code and data are publicly available (see Resources).

3. Results

3.1 Descriptive statistics

Across the 29 taboo words, mean sibilant density was 0.116 (SD = 0.152). Across the 29 matched neutral words, mean sibilant density was 0.076 (SD = 0.128). The taboo group thus showed approximately 52% higher sibilant density than the matched neutral group in absolute terms. Critically, the control variables were nearly identical between groups (Table 1).

Table 1. Group statistics for the matched pairs (*n* = 29 per group).

Statistic	Taboo group	Neutral group
Mean sibilant density	0.116	0.076
SD of sibilant density	0.152	0.128
Mean phoneme length	3.86	3.86
Mean log-frequency	2.83	2.82

3.2 Inferential statistics

The Shapiro-Wilk test indicated strong deviations from normality in both groups (taboo: $W = 0.748$, $p < 0.0001$; neutral: $W = 0.606$, $p < 0.0001$), which was expected given that sibilant density is a bounded measure with many zero values. The Wilcoxon signed-rank test was therefore taken as the primary inferential test.

The Wilcoxon signed-rank test, with the directional alternative hypothesis that taboo sibilant density exceeds neutral sibilant density, yielded $W = 51.5$, one-tailed $p = 0.16$. The paired t-test, reported for parametric comparison, yielded $t(28) = 1.17$, two-tailed $p = 0.254$, one-tailed $p = 0.127$. The paired Cohen's d was 0.22, conventionally interpreted as a small effect.

The trend in sibilant density was in the predicted direction, but the difference did not reach statistical significance at the conventional $\alpha = 0.05$ level. The null hypothesis cannot be rejected.

3.3 Visualization

Figure 1 displays the distribution of sibilant density values for both groups. The left panel shows boxplots with overlaid individual data points; the right panel shows histograms of the same data with dashed lines indicating group means. The two distributions visibly differ in their means but overlap substantially, which provides an intuitive explanation for the non-significant test result.

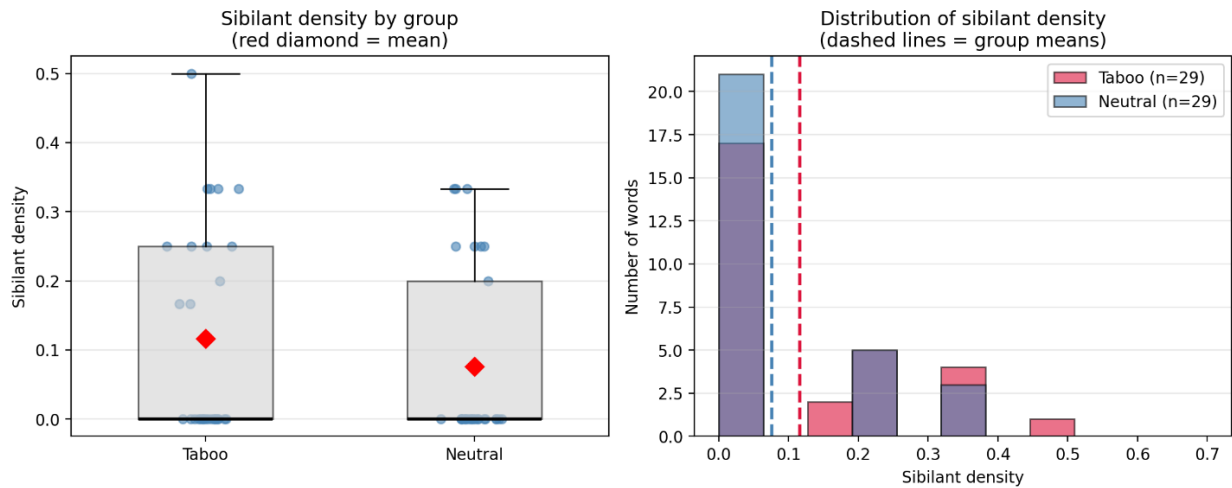


Figure 1. Sibilant density distribution for taboo ($n = 29$) and matched neutral ($n = 29$) words. Left panel: boxplots with individual data points overlaid (jittered) and group means marked as red diamonds. Right panel: overlapping histograms with dashed vertical lines indicating each group's mean. Although the group means differ in the predicted direction, the two distributions overlap substantially.

4. Discussion

The present study tested whether English taboo words are denser in sibilant phonemes than matched neutral words, a prediction motivated by Aryani et al.'s (2018) finding that hissing sibilants are perceived as more arousing and more negatively valenced. The descriptive results were consistent with the hypothesis - taboo words showed 52% higher sibilant density than matched neutral words, but the formal statistical tests did not reach significance at the conventional $\alpha = 0.05$ threshold. The null hypothesis cannot be rejected. Two substantive interpretations of this result are worth considering. The first is that sibilants exert only a modest lexical-statistical effect even though their acoustic-perceptual effect, as documented by Aryani et al., is robust. In other words, the perceptual prominence of sibilants in particular taboo words like *piss* or *shit* may not translate into a strong global statistical regularity over the entire taboo vocabulary. Inspection of the data supports this reading: many of the most clearly taboo items in English: *fuck*, *dick*, *cock*, *prick*, *cum*, *cunt*, contain no sibilants at all, while their phonological identity is carried instead by voiceless stops. The second interpretation, suggested by the same observation, is that the true phonological signature of taboo language may be broader than

sibilants alone, extending to voiceless stops and other "fortis" consonants that share certain acoustic properties (high intensity, sharp release). This is consistent with Aryani et al.'s broader finding that voiceless consonants generally drive perceived arousal. A future study could test this expanded hypothesis by computing density measures for sibilants combined with voiceless stops.

Several limitations of the present study should be acknowledged. First, the sample is small ($n = 29$ per group), and a power analysis based on the observed effect size suggests that approximately 130 matched pairs would be required to achieve 80% power at $\alpha = 0.05$, one-tailed. The study is therefore underpowered for the small effect that is present in the data, and a replication with a larger taboo word list is warranted. Second, sibilants were treated here as a binary, count-based category; a finer acoustic measure (e.g., spectral center of gravity weighted by phoneme duration) might reveal patterns invisible to the present analysis. Third, only American English pronunciation was used (via CMUdict), and the analysis used the first listed pronunciation variant for each word; dialect differences and pronunciation variability were not modeled. Fourth, the analysis was based on the lexical form of words and did not account for actual frequency of taboo use in spoken or written corpora, which may differ from the SUBTLEX subtitle frequencies.

Despite these limitations, the present study makes a methodologically clean contribution to a literature that has so far relied largely on perceptual ratings of pseudowords or on small-scale analyses of specific items. By using a frequency- and length-matched comparison group drawn from peer-reviewed affective norms, the study demonstrates that the directional sibilant-taboo association is real in the data, even if it is too small to confirm with the present sample size. Future work could combine larger taboo corpora (e.g., from Sulpizio et al., 2024), include voiceless stops as a complementary phonological category, and extend the analysis cross-linguistically to test whether the sound–taboo association generalizes beyond English.

5. References

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Data and code availability

All analysis code, source data files, and derived results are publicly available at:

<https://github.com/barcie10/sibilants-in-taboo-words>